

Demonstration of PDM-16QAM Transmission Robust to Fast SOP Variations Using Real-Time 2×2 MIMO Equalization on a GPU

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Abstract We demonstrated 5-Gbaud PDM-16QAM transmission using GPU-based real-time DSP with 2×2 MIMO equalization tailored for parallel processing. The pre-FEC BER remained stable below 10^{-4} for over one hour after 240-km SMF transmission, and degradation was small under fast SOP variations up to 10 krad/s. ©2026 The Author(s)

Introduction

Coherent detection and digital signal processing (DSP) have enabled high spectral efficiency through polarization-division multiplexing and higher-order modulation formats, as well as flexible compensation for various impairments [1–3]. In optical fiber transmission systems, DSP has typically been implemented using application-specific integrated circuits (ASICs) [4–6].

Recently, driven by the expansion of artificial intelligence technologies, the computational resources of graphics processing units (GPUs) have improved rapidly and been deployed on a large scale [7], making GPUs an attractive platform for flexible and upgradeable communications DSP [8]. In optical fiber communication systems, GPU-based real-time DSP has been demonstrated, as shown in Fig. 1 [9–15], including coherent detection [10] and Kramers-Kronig (KK) detection of single-polarization signals [11–13]. Real-time 20×2 multi-input multi-output (MIMO) equalization has also been demonstrated for 10-mode multiplexed transmission of 1-Gbaud QPSK signals [15]. However, the symbol rates in these demonstrations were limited to 1–2.5 Gbaud, whereas symbol rates in current systems can reach tens of Gbaud or beyond. Moreover, most of these demonstrations used single-polarization signals, and tolerance to fast state-of-polarization (SOP) variations was not evaluated [9–15]. Since real-time operation of GPU-based DSP relies on massive parallelization, fast SOP transients that may occur in practical environments [16, 17] can challenge adaptive MIMO convergence and tracking.

In this study, we demonstrated polarization-division-multiplexed (PDM)-16 quadrature amplitude modulation (QAM) transmission at 5 Gbaud using real-time DSP with 2×2 MIMO equalization on a GPU workstation. After 240-km transmission over single-mode fiber (SMF), the pre-forward error correction (FEC) bit error rate (BER) was below 10^{-4} and remained stable for over one hour. By adopting a DSP architecture tailored for parallel

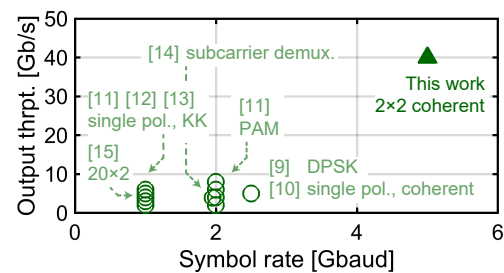


Fig. 1: DSP output throughput (total gross rate) and symbol rate of GPU-based real-time DSP demonstrations.

processing on the GPU, tolerance to SOP variations up to 10 krad/s was confirmed.

Real-time DSP implemented on a GPU

The experimental setup for transmission of 5-Gbaud PDM-16QAM using GPU-based real-time DSP is shown in Fig. 2. The GPU-based platform for real-time DSP consists of an optical front-end, digital-to-analog converters (DACs)/analog-to-digital converters (ADCs), a field-programmable gate array (FPGA) for ADC-GPU data transfer, and a GPU workstation. The sampling rates of the DACs and ADCs were 20 GS/s. The transmitted data consisted of FEC frames of the low-density parity-check code for DVB-S2, with a frame length of 64,800 and a code rate of 0.8, where the payload bits were random. Two FEC frames for each I/Q component of the X/Y polarizations were mapped to PDM-16QAM. Due to the DAC pattern length limit, a preamble of 52 symbols was added. A pilot symbol was inserted every 32 symbols for carrier recovery. Both the preamble and pilots consisted of random 16QAM symbols, with a total overhead of 3.2%. This resulted in a DAC pattern length of 66,944 symbols.

On the receiver side, the ADC outputs were downsampled to 10 GS/s ($2 \times$ oversampling) and rounded to six bits in the FPGA due to the limited communication bandwidth between the ADC and the GPU. The ADC data (240 Gb/s) was written directly into GPU memory on the GPU workstation via remote direct memory access (RDMA)

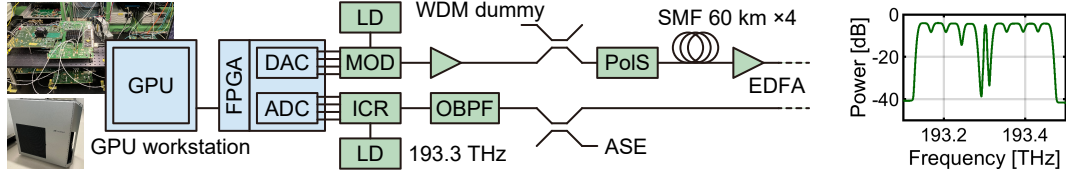


Fig. 2: Experimental setup for 5-Gbaud PDM-16QAM transmission using real-time DSP on a GPU workstation over 240 km SMF. The inset graph shows the transmitted optical spectrum with a resolution bandwidth of 0.02 nm. MOD: dual-polarization IQ modulator, ICR: integrated polarization-diversity coherent receiver, PolS: polarization scrambler, OBPF: optical bandpass filter.

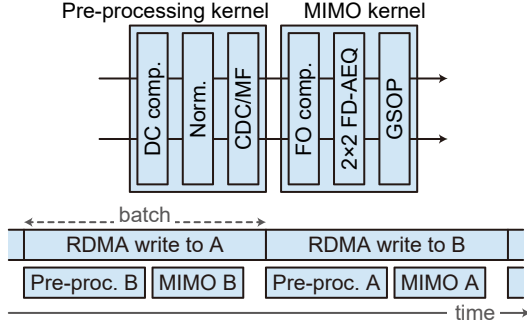


Fig. 3: DSP implemented on a GPU and real-time operation.

over Converged Ethernet (RoCE) v2. The GPU workstation was equipped with a 400 GbE network interface (Broadcom BCM957608-P1400GDF00), a CPU (Intel Xeon w5-2545), and an NVIDIA RTX PRO 6000 Blackwell Workstation Edition GPU.

The DSP implemented on the GPU is shown in Fig. 3. The Pre-processing kernel performed direct-current (DC) offset compensation, normalization, and fixed frequency-domain (FD) filtering with a fast Fourier transform (FFT)/inverse FFT (IFFT) size of 1024 and 50% overlap for chromatic dispersion compensation (CDC) and matched filtering (MF). The MIMO kernel performed frequency offset (FO) compensation, 2×2 FD adaptive equalization (AEQ) whose outputs were down-sampled to $1 \times$ sampling including carrier recovery [18], and Gram-Schmidt orthogonalization procedure (GSOP) [19]. The 2×2 FD adaptive filter was based on an overlap-save with FFT/IFFT sizes of 128/64 and 50% overlap. The coefficients were adaptively updated by data-aided/decision-directed least mean squares (LMS) in an unconstrained manner [20]. Carrier recovery was based on pilot symbols every 32 symbols and linear interpolation to allow for parallel processing. The phase error was estimated using pilot symbols, and a moving average with bias correction for the FO was applied. The MIMO kernel also includes BER evaluation.

Real-time DSP on the GPU was executed in a batch process, as shown in Fig. 3, in which one batch corresponded to 16,384 DAC patterns (approximately 10^9 symbol periods and 6.6 GB). If execution of the Pre-processing and MIMO kernels for one batch finishes within the batch duration, real-time operation without data loss is achieved.

High-throughput processing of the Pre-

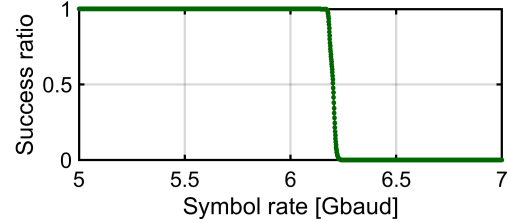


Fig. 4: On-time completion ratio of DSP consisting of Pre-processing and MIMO kernels on a GPU workstation.

processing and MIMO kernels requires massive parallelization. For the MIMO kernel, one batch of 16,384 patterns was divided into $N_{\text{parallel}} = 1024$ (corresponding to the GPU grid size) successive segments consisting of $N_{\text{pattern}} = 16$ patterns, and these segments were processed logically in parallel. The initial filter coefficients were the same across all parallel processes. Since inserting a training signal with a long period arbitrarily was not possible due to the limitation of the DACs, the transmitted pattern was regarded as a training sequence for the first 100,000 symbol periods (9.3%), and pre-convergence of the coefficients was performed using data-aided LMS before switching to decision-directed LMS and BER evaluation. Although data-aided pre-convergence cannot be eliminated with this parallelization scheme, its overhead decreases as available GPU memory increases. The Pre-processing kernel employed the same parallelization with $N_{\text{parallel}} = 256$ and $N_{\text{pattern}} = 64$.

We evaluated the DSP throughput on the GPU workstation by comparing the processing time of the Pre-processing and MIMO kernels required for one batch with the batch duration at each symbol rate. Figure 4 shows the on-time completion ratio over 10,000 trials. The GPU-based DSP completed within the required time up to 6.1 Gbaud. For higher symbol rates, the developed GPU-based DSP platform is limited by the communication bandwidth between the ADCs and the GPU and by the GPU processing throughput. Both limitations can be mitigated by employing multiple GPUs and multiple network interfaces to the ADC data, since TB/s-class high-speed inter-GPU communication, such as NVLink, is available [15, 21].

Evaluation results with real-time DSP on a GPU

We evaluated 5-Gbaud PDM-16QAM transmission over 240 km of SMF using GPU-based real-time

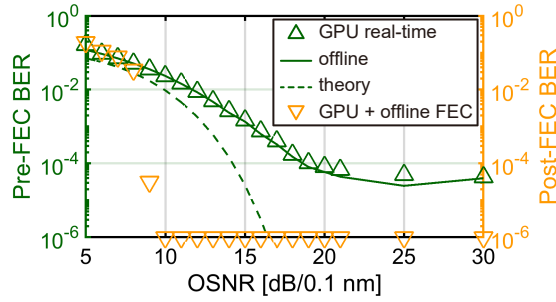


Fig. 5: Evaluation of OSNR sensitivity of 5-Gbaud PDM-16QAM with real-time DSP on a GPU in back-to-back configuration.

DSP. The 5-Gbaud PDM-16QAM signal was generated by the GPU-based DSP platform using a laser diode (LD) source at 193.3 THz with a linewidth below 2 kHz. The signal under evaluation was combined with eight neighboring wavelength-division multiplexed (WDM) channels, each with a 32-GHz rectangular spectrum spanning 193.15 to 193.45 THz on a 37.5-GHz grid, which were emulated by an amplified spontaneous emission (ASE) source and a wavelength-selective switch (WSS). The optical power of the 5-Gbaud signal was set to be 8 dB lower than that of the other eight 32-Gbaud signals to match the optical spectral power densities. The transmitted optical signal was then polarization-scrambled to emulate SOP variations.

The transmission link consisted of four 60-km SMF spans, totaling 240 km, and span losses were compensated by erbium-doped fiber amplifiers (EDFAs). The span input optical power was set to -15 dBm/GHz. The optical signal-to-noise ratio (OSNR) after 240 km was 28.5 dB/0.1 nm.

After the 240-km transmission, ASE noise was added to adjust the OSNR, and the signal at 193.3 THz was extracted and fed to the receiver side of the GPU-based DSP platform. The local oscillator had a linewidth below 2 kHz and was free-running relative to the carrier, with an FO of <20 MHz.

We first evaluated the OSNR sensitivity using the real-time DSP implemented on the GPU in a back-to-back configuration. The polarization scrambling speed was set to 10 rad/s. Figure 5 shows the averaged pre-FEC BER. We evaluated 400 consecutive batches (>1 min.) three times for each condition. For comparison, the theoretical curve [22] and the pre-FEC BER obtained by offline DSP on the signals input to the GPU from the FPGA were plotted. The post-FEC BER after offline FEC on a portion of the GPU-based DSP output was also shown. Although a pre-FEC BER floor appears at high OSNR due to imperfections in the devices and a limited vertical resolution of the GPU input, the pre-FEC BER after GPU-based real-time DSP was 2.4×10^{-2} at an OSNR of 10 dB/0.1 nm, and no bit errors were observed after offline FEC decoding.

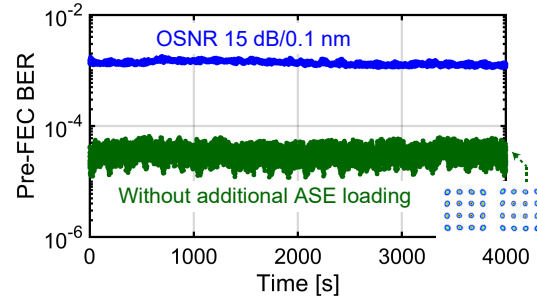


Fig. 6: Pre-FEC BER using GPU-based real-time DSP after 240 km, measured over more than one hour.

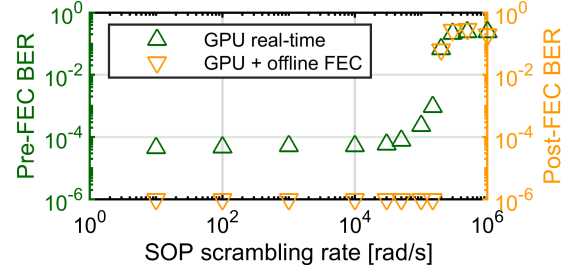


Fig. 7: Tolerance to SOP variations of GPU-based real-time DSP after 240-km transmission.

We next evaluated long-term real-time reception by the GPU-based DSP after 240-km transmission. Figure 6 shows the pre-FEC BER obtained with continuous GPU-based real-time DSP for more than one hour without additional ASE loading and with the received OSNR of 15 dB/0.1 nm. The long-term average pre-FEC BERs were 3.3×10^{-5} and 1.4×10^{-3} without additional ASE loading and with an OSNR of 15 dB/0.1 nm, respectively, and they remained stable for over one hour.

Finally, we evaluated the tolerance of the GPU-based real-time DSP to SOP variations by changing the polarization scrambling speed in the 240-km transmission. For each condition, 400 consecutive batches (>1 min.) were evaluated three times, and the average pre-FEC BER was plotted in Fig. 7. The post-FEC BER after offline FEC on the GPU-based DSP output was also shown. No significant degradation in the pre-FEC BER was observed as the scrambling rate was increased from 10 rad/s to 10 krad/s, confirming that the GPU-based real-time DSP was robust to SOP variations that can be on the order of kHz in practical environments [16, 17].

Conclusions

We demonstrated PDM-16QAM transmission at 5 Gbaud using real-time DSP with 2×2 MIMO equalization on a GPU workstation. After transmission over 240 km of SMF, the pre-FEC BER remained below 10^{-4} for over one hour, and tolerance to SOP variations up to 10 krad/s was confirmed. Regarding processing throughput, real-time operation was achieved at up to 6.1 Gbaud on the GPU workstation used in this experiment. Higher symbol rates may be feasible using multiple GPUs.

Acknowledgements

This work was partially supported by the National Institute of Information and Communications Technology, Japan (No. 24201). We thank Masaki Sato and Aridehide Noda for their support in the experiment, and John Kenji Clark for valuable comments on the manuscript.

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